

gram needs to handle. Objects may also represent user-defined data such as vectors, time and lists. When a program gets executed, the objects interact by sending suitable messages to one another. For example, any student will be an object in the class 'student'. Similarly, if we consider a fruit to be a class, then mango, apple, and guava will be objects in this class.

Dynamic Binding

The term binding refers to linking a procedure call to the code that needs to be executed as a response to this call. Similarly, 'Dynamic Binding' or late binding refers to the code that remains unknown until the procedure is called during run time. Dynamic binding is also associated with polymorphism and inheritance. For example, let us consider a procedure called 'calculation' declared in a class. Some classes may inherit that class. The definitions (code) associated with the procedure 'calculation' are written such that they perform different operations in each derived class. For example, in one class, the code is written for addition, while in another class for subtraction, and so on and so forth. For objects of different classes, the procedure will provide different results and will be unknown till the execution is complete.

Message Passing

Message passing is the process by which one object sends data to another, or asks the other object to invoke a method. This concept is also known as interfacing in some programming languages. In an object-oriented language, objects communicate with each other by sending and receiving various types of information. The message for an object is related to the request for execution of a procedure, and thus invokes a function in the receiving object, thereby generating the desired result.

Message passing also involves specifying the name of the object, the name of the message and the information to be sent. For example,